

Structural reductions toward the quartic Hessian conjecture in dimension four

Five-support exclusion, tangent nullcone containment, and exact secant colon certificates

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Abstract

We give a computer-assisted package of exact structural reductions for a quartic Hessian problem in dimension four. The package does **not** prove the quartic Hessian conjecture or the two-dimensional Jacobian conjecture. Its purpose is to isolate, certify and sharply reduce one repeated-factor boundary that had remained opaque in a direct normal-form programme.

There are four principal results. First, for a clean double-conic normal layer, a non-zero binary-decimic restriction with exactly five distinct projective roots cannot occur; the seven multiplicity partitions of ten into five positive parts are treated by exact characteristic-zero certificates. Second, on the tangent residual-quadratic orbit, the full 31-generator ideal of the binary-decimic multiplicity-six locus lies in the radical of the clean normal-layer ideal. Third, on the remaining secant chart, a $\mathbb{Z}/12$ -graded homogeneous reduction yields three successive exact localized-colon identities

$$Mh \in I, \quad Mh_2 \in (I, h), \quad Mh_3 \in (I, h, h_2),$$

over $\mathbb{Q}$. These are colon elements, not a computation of the full colon or saturation. Fourth, a prospectively selected batch of sixteen minimum-source-support free columns in the next Macaulay block lifts through 173^4 , reconstructs uniquely, and gives sixteen exact rational source-module syzygies. A separate audit expands all sixteen combinations to the zero polynomial over $\mathbb{Q}$.

The release includes frozen claim and metrics ledgers, exact receipts, reconstruction scripts, a 26 MB integer matrix interface, and fail-closed negative results. In particular, a fixed-gauge fourth target lifts through 173^{96} , but a preregistered exact-recovery hierarchy does not produce a rational target identity. The remaining secant-orbit containment, fourth target membership, colon equality, saturation and global lifting gates are stated explicitly.

1. Introduction

The Hessian-nilpotent reformulation of the Jacobian conjecture turns a global invertibility question into a collection of algebraic constraints on a homogeneous potential and its Hessian. Zhao showed that the Jacobian conjecture is equivalent to a vanishing conjecture for homogeneous quartic

Hessian-nilpotent polynomials (Zhao 2004); van den Essen and Zhao developed further vanishing criteria (Essen and Zhao 2007). Dimension three is known for gradient maps and constant Hessian determinant problems (Bondt 2012). The dimension-four quartic boundary remains a natural first place where repeated factors, non-trivial Hessian kernels and invariant theory interact.

This paper reports a structural-reductions campaign inside one such dimension-four programme. The starting point is a pinned normal-layer construction in Roy van Rijn’s public repository (<https://github.com/royvanrijn/jacobian-research>) at commit `3ed4544e8bbd9f2345c43612d1f3cf94fe279dc9`. The present package separately reconstructs the relevant covariant normal layers, then subjects the resulting systems to exact algebraic tests. Source reconstruction and new claims are kept separate throughout.

The central geometric object is a ternary quintic h_5 restricted to the Veronese conic

$$q = xz - y^2 = 0, \quad (x, y, z) = (s^2, st, t^2).$$

The restriction is a binary decimic f_{10} . A clean residual-line target has

$$\det \text{Hess}(h_5) = q^4 \ell \tag{1.1}$$

for a non-zero linear form ℓ . The repeated-factor boundary is therefore controlled by the root support of f_{10} and, on the stable locus, by the invariant theory of binary decimics.

The results should be read at three distinct levels:

1. ordinary mathematical implications, such as the seven-partition exhaustion and the exact rational identities;
2. finite certificates and deterministic replay, which establish the stated algebraic equalities for the frozen encodings;
3. assurance claims, which remain internal and producer-coordinated unless an unaffiliated reconstruction is explicitly recorded.

These levels are not interchangeable. A finite-field unit is not a characteristic-zero theorem; a long p-adic lift is not rational reconstruction; an exact source syzygy is not membership of an inhomogeneous target; and internal review is not peer review.

1.1 Main results

The first result eliminates the exactly-five-support stratum.

Theorem A (five-support exclusion, computer-assisted). Let K be a field of characteristic zero. Let $h_5 \in K[x, y, z]$, and let its non-zero restriction $f_{10}(s, t) = h_5(s^2, st, t^2)$ have exactly five distinct roots in $\mathbb{P}^1(\overline{K})$. Then there is no non-zero linear form ℓ for which (1.1) holds.

The next result closes the tangent half of the stable nullcone target.

Theorem B (tangent normal-layer containment, computer-assisted). Let I_{tan} be the characteristic-zero clean normal-layer ideal after the tangent residual-quadratic substitution. Let N_6 be the defining ideal of the binary-decimic coincident-root locus

$$X_{(6,1,1,1,1)} = \{L^6 Q_4\} \subset \mathbb{P}(\mathrm{Sym}^{10} K^2).$$

Then

$$N_6 \subseteq \sqrt{I_{\mathrm{tan}}}. \quad (1.2)$$

Equivalently, every algebraic-closure point of the tangent normal-layer system has a binary-decimic restriction with a root of multiplicity at least six.

On the secant orbit, the package proves a smaller exact chain rather than closure.

Theorem C (three localized-colon elements). For the frozen remaining secant chart, let $I = (F_1, \dots, F_{17})$ be the homogeneous cubic ideal and let

$$M = f_9 f_{10} (2f_9^2 + 5f_{10} g_0).$$

There are explicit rational quartics h, h_2, h_3 and explicit rational multipliers satisfying

$$\begin{aligned} Mh &= \sum_{i=1}^{17} q_i F_i, \\ Mh_2 &= \sum_{i=1}^{17} r_i F_i + r_{18} h, \\ Mh_3 &= \sum_{i=1}^{17} s_i F_i + s_{18} h + s_{19} h_2. \end{aligned} \quad (1.3)$$

The three identities are verified coefficientwise over $\mathbb{Q}$. They show

$$h \in I : M, \quad h_2 \in (I, h) : M, \quad h_3 \in (I, h, h_2) : M.$$

They do not show that any displayed colon is generated by the exhibited element, and they do not prove $I : M^\infty = (1)$.

The fourth result is new to the closing experiment of this release.

Theorem D (sixteen rational fourth-block source syzygies, computer-assisted). In the frozen degree-eight, character-two fourth Macaulay source block, partition the 38,826 columns as $[B \mid C]$, where B consists of 36,587 characteristic-173 pivot columns. For the prospectively selected free coordinates

$$46, 47, 57, 66, 68, 102, 155, 547, 763, 965, 966, 1261, 1269, 1499, 1699, 1711, \quad (1.4)$$

let C_{16} be the corresponding source columns under the frozen primitive row scale. There exists an explicit matrix $T \in \mathrm{Mat}_{36587 \times 16}(\mathbb{Q})$ such that

$$BT + C_{16} = 0. \quad (1.5)$$

The first column of T has 36 non-zero entries and each remaining column has 21. After adjoining the distinguished unit in C_{16} , the sixteen source-module syzygies have supports 37 and $22^{\times 15}$, respectively.

Theorem D is a source-kernel theorem. It does not assert that the fourth inhomogeneous target lies in the image of B , nor that the rational kernel directions suffice to change the failed fixed gauge into a target solution.

1.2 What remains open

The following implications are *not* claimed:

- Theorem B does not transfer from the tangent residual orbit to the secant residual orbit.
- Theorem C does not determine the full colon or saturation.
- Theorem D does not imply rational membership of the fourth target Mh_4 .
- The five-support and tangent results do not close restrictions with six or more distinct support points.
- The clean normal-layer results do not by themselves prove polynomial-level orbit closure or the global quartic Hessian conjecture.

2. Binary decimics and the nullcone target

Write $V_{10} = \text{Sym}^{10}(K^2)$. The nullcone of the natural SL_2 -action consists of binary decimics with a root of multiplicity at least six. Brouwer and Popoviciu give a homogeneous system of parameters of degrees

$$2, 4, 6, 6, 8, 9, 10, 14$$

for the invariant ring (Brouwer and Popoviciu 2010). Their nullcone characterisation is classical input here, not a new result.

The same target is the coincident-root locus $X_{(6,1,1,1,1)}$. Equations and local presentations for multiple-root loci were developed by Chipalkatti (Chipalkatti 2003) and corrected and refined by Kurmann (Kurmann 2011); Gordan ideals provide another classical description (Weyman 1993). The conormal and duality geometry of multiple-root loci gives a further perspective (Lee and Sturmfels 2016).

An exact Macaulay2 construction in the present package gives the following profile.

Proposition 2.1 (target-ideal profile). The projective variety $X_{(6,1,1,1,1)}$ has dimension five, codimension five and degree 30. Its defining ideal has 31 minimal generators, decomposing under SL_2 as

degree	modules	dimension
2	V_0	1
3	$V_2 \oplus V_6$	$3 + 7$
4	$V_0 \oplus V_4 \oplus V_4 \oplus V_8$	$1 + 5 + 5 + 9$.

(2.1)

Thus an equivariant containment proof needs seven highest-weight families, not 31 unrelated coordinate calculations.

3. Exactly five support points

Assume f_{10} has exactly five distinct projective roots. After a projective change of binary variables, three roots may be placed at $0, \infty, 1$, while the other two have coordinates λ, μ . The distinct-root condition is

$$\Delta = \lambda\mu(\lambda - 1)(\mu - 1)(\lambda - \mu) \neq 0. \quad (3.1)$$

The multiplicities form a partition of ten into five positive parts. Up to ordering there are seven:

$$\begin{aligned} (6, 1, 1, 1, 1), (5, 2, 1, 1, 1), (4, 3, 1, 1, 1), (4, 2, 2, 1, 1), \\ (3, 3, 2, 1, 1), (3, 2, 2, 2, 1), (2, 2, 2, 2, 2). \end{aligned} \quad (3.2)$$

Every normal form is represented by

$$s^a t^b (s - t)^c (s - \lambda t)^d (s - \mu t)^e.$$

The induced Sym^2 -action preserves the conic up to a unit, so existence of (1.1) is invariant under the normalization. An arbitrary ternary quintic with the given restriction is the canonical harmonic lift plus qG_3 for an arbitrary ternary cubic G_3 .

For each partition the producer expands

$$D = \det \text{Hess}(h_5)$$

over $\mathbb{Q}$, extracts the residual coordinates

$$A = [x^5 z^4]D, \quad B = [x^4 y z^4]D, \quad C = [x^4 z^5]D,$$

and generates the 52 non-zero coefficients of

$$D - q^4(Ax + By + Cz). \quad (3.3)$$

Each row certificate proves

$$(A, B, C) \subseteq \sqrt{I K[\lambda, \mu, g_0, \dots, g_9, 1/\Delta]}. \quad (3.4)$$

The proofs use branch-complete radical chains, endpoint staircases and fixed-content identities. Whenever a pivot factor is inverted, its zero fibre is treated as a separate branch. No exceptional component is removed without a receipt. Since (3.2) is exhaustive, (3.4) proves Theorem A over $\mathbb{Q}$; the displayed rational identities and radical containments then persist after base change to any characteristic-zero field.

The divisor $\Delta = 0$ merges roots and therefore belongs to support at most four. The package records this geometric boundary separately. A further support-at-most-four corollary depends on the pinned source commit and is not promoted to a separately reconstructed result here.

4. Tangent-orbit nullcone containment

The residual quadratic has two non-zero SL_2 -orbits, tangent and secant. On the tangent orbit, a torus action normalises the quadratic invariant j_2 to one. A first-jet stratification indexed by the largest non-zero endpoint coefficient $r = 5, \dots, 10$ reduces the original presentation to six exact cases.

The $r = 5$ case is closed by a two-branch compatibility determinant; $r = 6, \dots, 9$ use coefficient-normalised exact standard bases; and $r = 10$ uses a convolution identity followed by five leading-degree branches. All branches are over characteristic zero.

For each of the seven irreducible families in (2.1), one highest-weight coordinate is shown to lie in $\sqrt{I_{\text{tan}}}$ by an exact unit-ideal cover. The tangent stabiliser preserves the normal-layer ideal. Exact orbit-rank calculations therefore span every coordinate of each family. The module dimensions add to $1 + 10 + 20 = 31$, proving (1.2).

This is an exact computer-assisted radical-containment theorem on one residual orbit. It is not a numerical sample and not a transfer from one highest weight without a stabiliser argument. It remains internal: no unaffiliated team has rebuilt the complete proof.

5. The secant obstruction and colon chain

The analogous secant j_2 -problem closes on first-jet strata $r = 5, \dots, 9$ and on all but one $r = 10$ branch. The remaining chart has $f_9 \neq 0$ and leading convolution degree six. Four triangular unit-linear relations eliminate g_4, g_5, g_7, g_8 , leaving 13 equations in 14 homogeneous variables. In the unreduced coordinates, the frozen homogeneous form is

$$I = (F_1, \dots, F_{17}) \subset \mathbb{Q}[f_3, \dots, f_{10}, g_0, g_1, g_2, g_3, g_6, g_9]$$

with open factor $M = f_9 f_{10} (2f_9^2 + 5f_{10}g_0)$. The unresolved geometric statement is

$$I : M^\infty = (1). \tag{5.1}$$

An exact Smith-form audit of within-generator exponent differences gives the universal grading quotient

$$\mathbb{Z} \oplus \mathbb{Z}/12. \tag{5.2}$$

The free factor is total degree, the auxiliary inverse has character zero, and M has character one. This grading isolates small degree-eight Macaulay blocks.

The three identities (1.3) were found by finite-field kernel extraction and rational reconstruction but certified by exact characteristic-zero replay. The first identity has 6,301 non-zero multiplier

coefficients, the second 15,211 and the third 19,050. For the third identity, a separate semantic audit rebuilds 19 generators, 38,048 multiplier descriptors and all 85,651 degree-eight monomial rows, obtaining zero coefficient mismatches.

Successive colon elements are evidence of structure: multiplication by M repeatedly returns new quartic quotient classes to the preceding ideal. They are not, however, a finite-generation theorem for the entire saturation. In particular, (5.1) does not follow from three examples.

6. Source kernels, quotient charts and failed routes

The third-colon source matrix contains an exact 2,053-dimensional rational Koszul subspace. A matching computation proves rank 2,053 over $\mathbb{Q}$, and a separate 72-digit lift gives an exact rational 114-dimensional quotient chart. Over $\mathbb{F}_{181}$, the Koszul block and a canonical residual section give a full $2,053 + 114$ kernel decomposition. Several sparse residual directions lift to exact rational syzygies.

Two negative routes are preserved because they delimit what the positive results do not show. A selected 2,053-square coordinate minor is singular in characteristic 197, so that fixed-minor normalisation stops fail-closed. A Python Dixon factorisation family exceeds its frozen wall/RSS gates before a factorisation artifact is produced. Neither failure is evidence against the underlying membership statement; each rejects only its registered architecture.

Compiled sparse LinBox elimination changes the computational scale. The actual 85,651 by 35,881 third-block coefficient matrix has full selected-column rank modulo 181, with the full rank computation taking 35.66 seconds. This calibrated capability motivates the multi-right-hand-side experiment in the next section.

7. The fourth target and the 16-column experiment

After adjoining h, h_2, h_3 , the next inhomogeneous target system has 85,688 rows, 36,587 selected pivot coordinates, 2,239 free coordinates and 1,487,624 selected non-zero entries. A fixed-gauge solution lifts through 173^{96} , and a separate audit replays every correction and the terminal congruence.

That long lift does not reconstruct in the frozen hierarchy. At height 96, equal-height reconstruction resolves 28,008 coordinates and leaves 8,579 unresolved. A stable-denominator candidate fails 42,870 exact rows, and fixed simultaneous-LLL blocks of dimensions 8, 16, 24 and 32 fail their sampled exact replays. This is a negative result about one gauge and one reconstruction family—not a proof of non-membership.

7.1 Prospective selection

Before computing a transfer, the 2,239 free source columns were ordered by

(source support, global coordinate).

Supports range from 13 to 248, and 43 columns have minimum support 13. The first sixteen in that fixed ordering are (1.4), all in the same generator family. No target vector was used to choose them.

Write the global source matrix as $[B \mid C]$. The experiment reuses the primitive scale of each existing B -row. The new C_{16} coefficients do not participate in a second row-content normalisation. Two of the 208 scaled source entries are rational, with denominators that are units at 173.

The exporter reconstructs the complete rational row stream and verifies the frozen generator, target, descriptor and monomial hashes. It then compares every primitive B -row against the 1,487,624-entry integer matrix. The observed mismatch count is zero.

7.2 Canonical section and p-adic lift

Over $\mathbb{F}_{173}$, the nullspace of $[B \mid C_{16}]$ has dimension 16. The lower 16 by 16 block is invertible, so right-normalisation produces

$$T_0 = -B^{-1}C_{16} \pmod{173}.$$

All 85,688 rows replay exactly. Three Hensel correction steps reuse the same B . At height k , the accumulated vector is checked modulo 173^k , the residual is divided by 173^k , and a shared 16-right-hand-side solve produces the next digit. Every augmented nullity is 16 and every replay mismatch count is zero.

The four LinBox solves take 108.999, 106.158, 113.824 and 110.353 seconds, respectively, for a total of 439.334 seconds. The first two use 215.157 seconds, below the prospectively frozen 460-second promotion ceiling. Peak observed solver RSS is about 426 MB, with zero swaps.

7.3 Rational reconstruction and implementation-diverse audit

At modulus

$$173^4 = 895745041,$$

the equal numerator/denominator uniqueness bound is 21,162. All 585,392 coordinates resolve. Exact rational replay of $BT + C_{16}$ has zero mismatches, proving (1.5).

The separate internal audit does not read the CSR rows. It maps each non-zero coordinate back to its generator and multiplier monomial, adjoins the selected free coefficient one, and expands the resulting source combination directly over $\mathbb{Q}$. Each of the sixteen residual polynomials has zero terms. This proves Theorem D without relying on the p-adic row encoding, subject to the shared frozen generator definitions.

8. Claim architecture and remaining gates

The contribution of each result to the quartic Hessian programme is summarised below.

unit	exact contribution	next missing implication
five-support theorem	excludes every clean exactly-five-root restriction	six-or-more support and full double-conic packet
tangent nullcone theorem	all 31 multiplicity-six generators vanish on the tangent normal layer	secant nullcone containment
secant colon chain	three exact new localized-colon elements	full colon or saturation
fourth target lift	certified congruence through 173 ⁹⁶	rational target identity
C16 theorem	sixteen exact rational source syzygies	prove they can move the target to an exactly reconstructible gauge
global programme	sharper clean normal-layer frontier	polynomial-level lifting and HC4

The most tractable successor is now a **gauge-action test**, not another blind precision extension: apply the exact 16-dimensional rational kernel family to the fourth target, derive the height objective explicitly, and test whether a rationally simple target representative exists. That successor requires its own preregistration because Theorem D proves kernel freedom but does not select or certify an optimal gauge.

9. Reproducibility and assurance

The repository contains:

- the manuscript in Markdown and LaTeX;
- CLAIM_LEDGER.json, with theorem, bounded-certificate and open-gate labels;
- RESEARCH_METRICS.md, separating construction, solve, audit and failed-route timings;
- all campaign receipts and scripts needed by the included replay paths;
- the frozen integer and characteristic-173 B -matrix interfaces;
- the C16 p-adic digits, rational candidates and terminal receipt;
- the direct-polynomial implementation-diverse audit receipt;
- the pinned-source boundary and novelty-gate report.

The public research repository is <https://github.com/ipitchford/hc4-five-support-structural-reductions>, and the version DOI is <https://doi.org/10.5281/zenodo.22216400>.

The principal closing receipts are:

651b952e7a228014b91d3ef28eb3954b7ba2265c2d0de302390b27e77c25eb25	p4-reconstruction.json
0f66ed6bd7182ffda4f7e4f34deca1eb80e785488b88fa66182aacd00bafd8cd	rational-candidates.json
ace9072d6e00b305350d617901e6a134fe52b8905d5447c4a6db5a14171ca5d7	direct-polynomial audit

The result is an anonymous, AI-assisted, unrefereed candidate. The internal audit uses a separate direct-polynomial implementation, but it is still producer-coordinated and does not count as unaffiliated reproduction. Independent rerun, independent reimplementations, formal verification, authenticated specialist review, editorial peer review and priority assessment remain absent or partial as stated in the machine record.

10. Conclusion

The repeated-factor HC4 boundary is more structured than the original large systems suggested. Exactly five support points are excluded; the tangent residual orbit lies in the multiplicity-six nullcone; the remaining secant chart carries an exact three-step colon chain; and the fourth source block has a sparse rational 16-dimensional kernel slice found by a prospectively frozen p-adic experiment.

The decisive missing statement is still not cosmetic. Neither the secant saturation nor rational fourth-target membership has been proved. The value of the present package is therefore a changed frontier: a diffuse family of large symbolic systems has been replaced by exact orbitwise statements, explicit identities, reusable rational syzygies and sharply stated next gates.

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